from the one Stevin had introduced in the beginning of the seventeenth century (figure VI.38). Before presenting his own instrument, Hoffmann gave a cursory introduction to perspective. He also published books on how to draw and paint and on colours. In 1799 he was appointed *Zeichenmeister* (master of drawing) in Erlangen.

Bischoff and Bürja

The three eighteenth-century mathematicians Bischoff, Lambert, and Bürja each invented and described a perspective instrument (Bischoff 1741, Lambert 1752, Bürja 1795). Bürja's presentation occurred in a book on perspective, to which I come back. On perspective machines in general, he had the following comment.

> I have noticed various troubles and inconveniences in the use of those perspective instruments with which I have become acquainted.[3]

This statement confirms my impression, indicated earlier, that perspective instruments were better as creations of the mind than as practical tools. Bürja's own instrument resembles Scheiner's pantograph, and it is difficult to say whether it really was convenient to use. I return in chapter XII to Lambert's instrument, but find it appropriate to mention here that it is also doubtful whether Lambert's machine would have worked smoothly (page 643).

Johann Christoph Bischoff (†1774) described his instrument in *Kurtzgefasste Einleitung zur Perspectiv, darinnen nebst dem wahren Fundamente derselben ...* (Brief introduction to perspective containing its true foundation together with ..., 1741). Unfortunately I can say nothing about his instrument, because I have only been able to examine a copy of Bischoff's book in which the plates were missing. In treating perspective, Bischoff based his presentation on a distance point construction that he introduced without including any theory – despite the title of his book. In his preface, he wrote that his reason for writing yet another book on perspective was that many of the existing works could only be accessed with difficulty, if at all.

XI.3 Anamorphoses

Albrecht

The diagram reproduced in figure XI. 5 shows that a kind of distance point method for constructing anamorphoses was already applied in Germany in the mid-sixteenth century. As we saw in chapter IX, de Caus

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[3] *Bei denjeningen perspektivischen Instrumenten, die zu meiner Kenntniss gekommen sind, habe ich allemal in der Anwendung verschiedene Schwierigkeiten und Unbequemlichkeiten bemerket.* [Bürja 1795, 225]

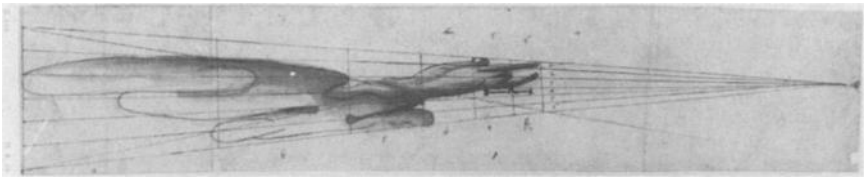

FIGURE XI.5. A diagonal method applied for an anamorphic drawing. The author is known as Master H.R. from Nuremberg, c. 1540. Universitätsbibliothek Erlangen-Nürnberg drawing B 448.

described such a method in 1612, though not very understandably – the first lucid exposition in French being Niceron's presentation from 1638. Between the appearances of these two works, Albrecht published the first German description of how to construct a perspectival anamorphosis, entitled *Zwey Bücher. Das erste von der ohne und durch die Arithmetica gefundenen Perspectiva. Das andere von dem darzu gehörigen Schatten* (Two books. The first on perspective found without and by means of arithmetic. The other about the shadow belonging to it), published in 1623 (figure XI.6). Before telling any more about Albrecht's anamorphoses, let me present the author and his book shedding light on its rather enigmatic title.

Andreas Albrecht (†1628) was as a mathematical practitioner earning his daily bread as a calculator and engineer. He evidently did not tire easily of arithmetical operations, because he devoted a large part of his work on perspective to calculations. Like Dürer, Albrecht centred his work around a plan and elevation construction of the perspective image of a cube with two sides parallel to the picture plane π. Unlike Dürer, however, Albrecht assumed that the eye point is situated centrally with respect to the cube – or more precisely, that the eye point lies in the vertical plane orthogonal to π that bisects the cube.

Having presented a geometrical construction of the image of the cube, Albrecht proceeded with calculations. He chose a fixed *distance* and made some tables listing the coordinates (though he did not use this term) of the perspective image of a vertex in the cube for various values of the height of the eye point above the ground plane, the distance between the cube and the picture plane, and the side of the cube (figure XI.7). He stressed that in choosing the various variables, it was important to be aware that the resulting visual angle should have a reasonable size under 90° (Albrecht 1623, chapter one).

Thus far we have not met an explicitly arithmetical approach to perspective, by which I mean an approach in which an author recommends his reader to calculate what corresponds to the coordinates of a perspective image. This is understandable, since before the age of modern computers it was, as noted previously, much quicker to find the image using a geometrical construction than it was to calculate its coordinates. However, we have seen methods that implicitly involve arithmetic, namely the procedures that

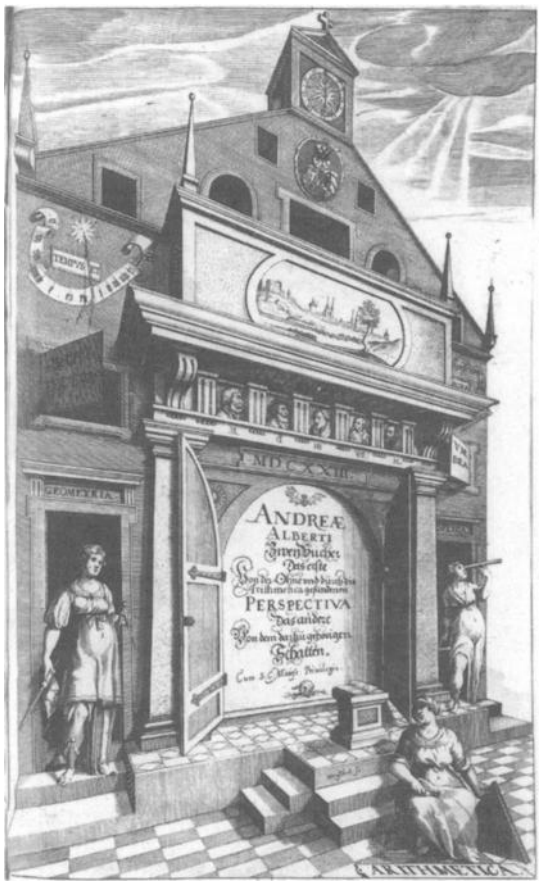

FIGURE XI.6. The title page of Albrecht's book.

were based on the use of special scales – often engraved on a sector. This solution was one that was preferred, most notably, by a number of French perspectivists from around 1630 onwards (while others constructed foreshortening scales geometrically). Albrecht's procedure of producing tables rather than scales is quite exceptional; one of the few others to use this approach was Claude de Roy (page 477), who presumably did this completely independently of Albrecht.

As indicated in the title of Albrecht's book, he also treated shadows. He included a few more examples than Dürer, but basically followed Dürer and like him, drew a sun as the source of light, but still treated the source as a point. In extension of his treatment of shadows, Albrecht turned to an example of an anamorphosis, which he called an *Ongestalt*.[4] He applied a plan and eleva-

[4]In modern German: *Ungestalt* (deformed object).

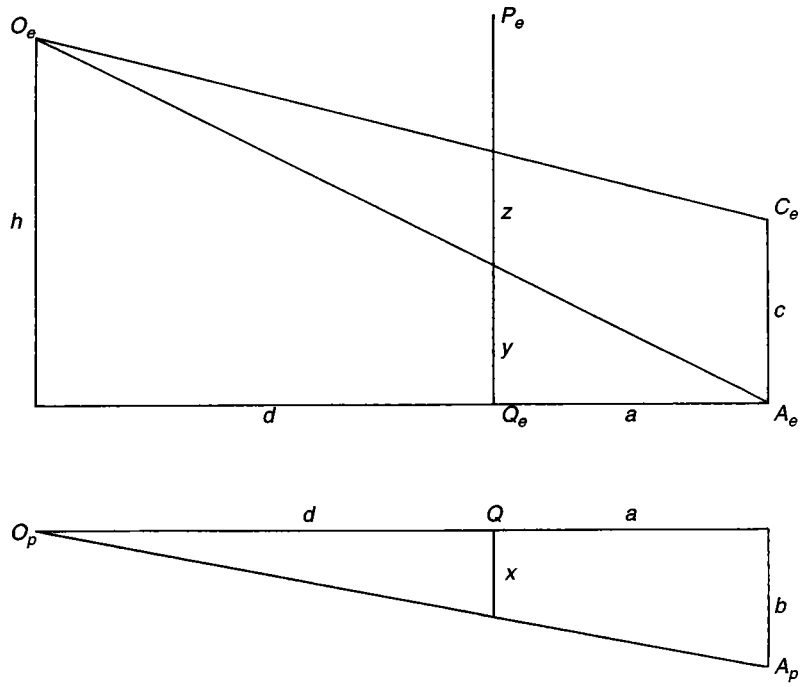

FIGURE XI.7. Diagram to Albrecht's tables. The upper drawing shows the elevations O_e, A_e, and C_e of the eye point O, a point A in the ground plane, and a point C vertically above A – the picture plane π being represented by P_eQ_e. Similarly, in the lower drawing O_p and A_p are the projections of O and A into the plan. Moreover, d is the *distance*, h the height of O above the plan, a the distance between A and the picture plane, b the distance between A and the *vertical plane*, and c the distance between C and the plan. Finally x, y, and z are the foreshortened b, a, and c respectively. Albrecht's arithmetical approach to perspective consisted in calculating tables, in which x, y, and z could be looked up.

From similar triangles we see that the required quantities are given by the following relations,

$$x = \frac{bd}{a+d}, y = \frac{ah}{a+d}, \text{ and } z = \frac{cd}{a+d}$$

Albrecht set d equal to 10 and let a, c, and h be integers in the interval [1,16] and b be integers in the interval [1,8]. He first used the rule of three to find x, y, and z, but later he also introduced some angles and their tangents, and obtained some of his results in a rather cumbersome way. With his method of calculating x, y, and z Albrecht seems to have confused himself, for he did not notice that z is independent of h.

tion technique in a correct and straightforward manner to construct the anamorphosis. The caption of figure XI.8 explains the principle of Albrecht's construction, and his own illustration is reproduced in figure XI. 9.

Besides his very detailed explanation of how a perspective cube can be obtained in various ways, Albrecht included a number of illustrations of

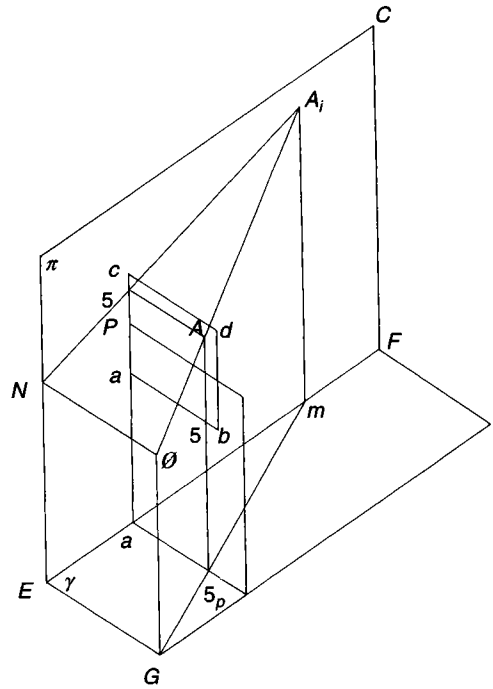

FIGURE XI.8. The principle behind Albrecht's construction of anamorphoses. To coordinate this diagram with the one by Albrecht, reproduced in the next figure, I have used his letters – and added a few of my own. The square *abcd* is to be projected upon the wall π from the eye point *Ø*. Albrecth did this with the help of the square's plan and elevation, where, as plan he used the ground plane γ and as elevation π itself, *EF* being the line of intersection of these two planes, and *G* and *N* the plan and elevation of *Ø*. In the diagram I have showed how he constructed the image of the point *A* (5,5): The plan and the elevation of the line *ØA* are the lines $G5_p$ and *N5*. Using a traditional plan and elevation construction, Albrecht found the point of intersection *m* of *EF* and $G5_p$, and finally determined the image A_i as the point of intersection of *N5* and the vertical line through *m*.

elaborate perspective compositions, among them a spiral staircase, which became a much-used motif among Dürer's followers. As the first seventeenth-century German work completely devoted to perspective, Albrecht's contribution seems to have become fairly well known. It was translated into Latin,[5] and was often referred to by later writers. However, it was more the existence of Albrecht's work than its content that was mentioned. His treatment of anamorphoses seems to have been particularly overlooked: the subject was taken up by several other German writers, but they used Niceron's method.

Kircher and Schott

In chapter IX we saw how Niceron's interest in anamorphoses was linked to a general enthusiasm for the aspects of science that can seem magical. A similar attitude towards science was displayed by the Jesuit polymath Athanasius Kircher (c. 1602–1680) who had contacts with the Minims – the Order to which Niceron belonged. Kircher was attracted by virtually all the

[5]The Latin edition I have seen was partially in German, but whether this applies to all copies, I cannot say.

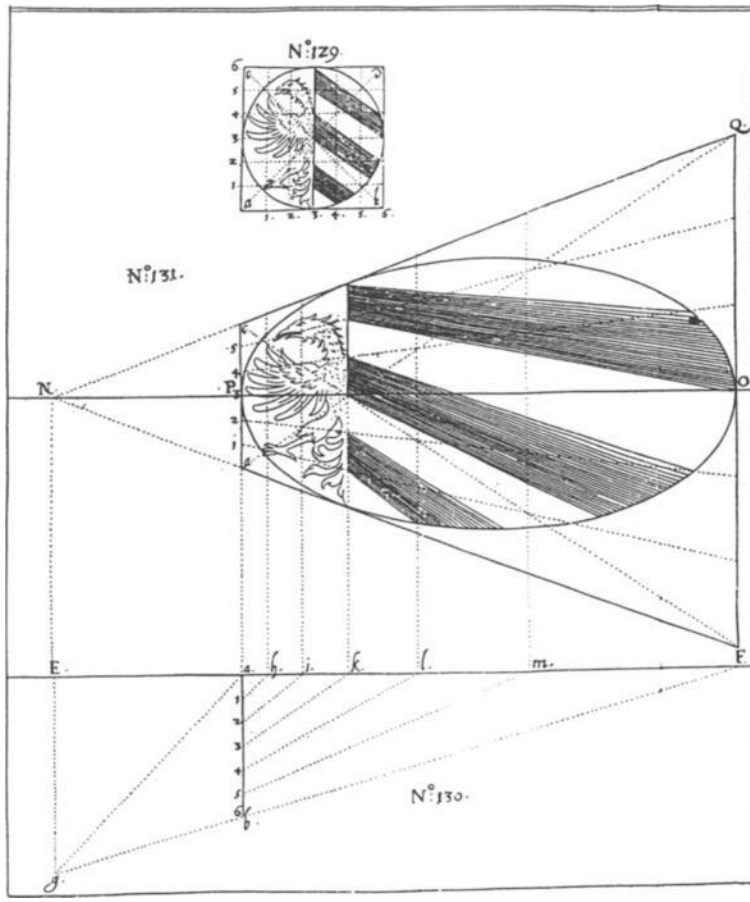

FIGURE XI.9. Albrecht's construction of an anamorphosis of a Nuremberg eagle. Albrecht 1623, figures 129–131.

theoretical disciplines of his time within science and the humanities – including hieroglyph deciphering (Kangro[S] 1973). Moreover, he was very eager to transmit his knowledge and wrote more than forty books. Kircher not only studied curiosities, but also collected them. During a prolonged stay in Rome he collected a number of strange artifacts to a museum, which, although not created by him, came to be known as Kircher's museum. The museum is likely to have contained some anamorphoses, for Kircher took a vivid interest in them (for more on Kircher and anamorphoses – and other curious pictures – see Baltrušaitis[S] 1977, 79–86).

In 1646 Kircher published *Ars magna lucis et umbrae* (The great art of light and shadow), which dealt with various disciplines involving light rays and included perspective and anamorphoses (Kircher 1646). He used a number of uncommon terms, for instance *planum mesopticum* for the picture plane and rather confusingly, *punctum oculi* (eye point) for a distance point. He presented a few mathematical results such as the convergence rule – but did not introduce the general concept of a vanishing point. As his method of construction he chose a distance point construction, which he also applied for perspectival anamorphoses. His treatment of various sorts of anamorphoses seems to be inspired by Niceron's work. For cylindrical mirror anamorphoses, Kircher presented the same construction as Niceron (figure IX.38), and like Niceron, he suggested that the radii of two consecutive circles should have the ratio 21:20.

Kircher exerted a great influence on his pupil Gaspar Schott (1608–1666), who also entered the Jesuit Order. Like his master, Schott was much fascinated by curious phenomena – natural as well as supernatural. Today he is best known for his role as a correspondent who kept himself and others abreast of various experiments, in particular some concerning vacuum (Keller[S] 1975, 210–211). Schott wrote about his observations in *Magia universalis naturae et artis* (Universal magic of nature and art). In this book he included optics and devoted seventy pages to the "magic of anamorphoses" (Schott 1657). He followed Kircher in presenting perspective in connection with anamorphoses (for more on Schott and anamorphoses, see Baltrušaitis[S] 1977, 85–90). As we shall see, Schott also took up perspective in a textbook on mathematics that appeared posthumously.

Leupold

After the 1650s anamorphoses apparently lost some of their attraction – and presumably went out of fashion. They continued to be treated in books on perspective, but did not receive much attention. The golden age of anamorphoses partially overlaps with the period when perspective instruments were greatly in focus. One could imagine that during this time some people got the idea of creating machines that could, at least in principle, construct anamorphoses. I do not know whether this really happened at the time, but such instruments were described later, by the instrument maker Jakob Leupold. In 1713 Leupold published *Anamorphosis mechanica nova, oder Beschreibung dreyer neuen Maschinen* (New machines for anamorphoses, or description of three new machines). It is a kind of catalogue consisting of six pages that describe, indistinctly, three rather complicated machines, all meant for constructing mirror anamorphoses: one for plane mirrors, one for conical mirrors, and one for cylindrical mirrors (figure XI.10). For further reading Leupold recommended, among others, Niceron, Kircher, Schott, and Sturm.

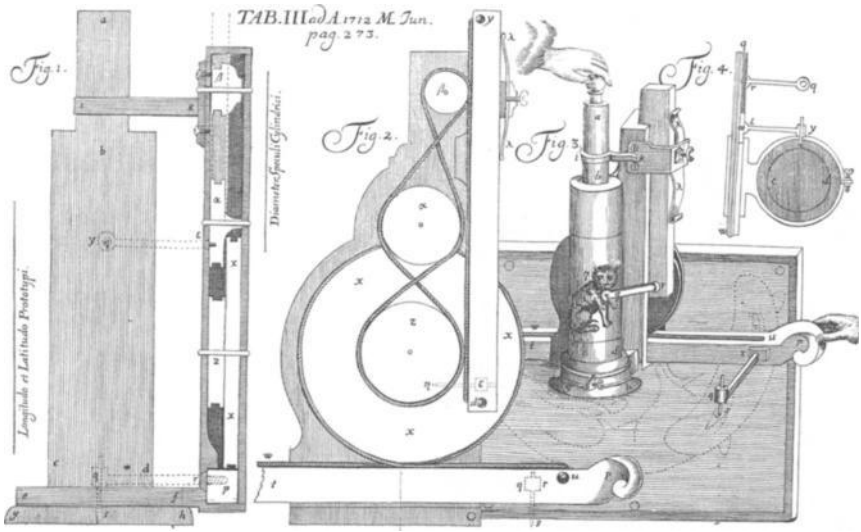

FIGURE XI.10. Leupold's machine for creating cylindrical anamorphoses. Contrary to my general policy, here I abstain from describing the machine's mode of operation, but I can tell as much as the drawing fixed to the cylinder in *fig. 3* is redrawn as an anamorphosis on the horizontal board under the cylinder. Leupold 1713, being a reproduction of a figure published a year earlier in *Acta eruditorum*.

Mathematischer Lust und Nutzgarten

With the only indication that it was written by "S.R.", a book appeared in 1724 under the title *Mathematischer Lust und Nutzgarten* (The mathematical pleasure and vegetable garden). Among many other things, this work treated anamorphoses and carried a brief introduction to perspective.

The search to identify S.R. has been a puzzle, but I am now convinced that two authors were involved, namely the artist Johann Jacob Schübler (1689–1741) – to whom I return – and the astronomer Johann Leonhard Rost (1688–1727). To support my conclusion I first relied on the following two arguments. First, one of the older biographies of Schübler mentioned his being the author of a *Mathematischer Lustgarten* (Will^{S} 1757). Second, in 1745, after both men had died, another *Mathematischer Lust und Nutzgarten* was issued with Rost's name on the title page and with the rest of the title similar to that of the S. R. edition (Rost 1745).[6] Later I found an irresistible

[6]I cannot tell whether Schübler's name also figures in 1745, nor compare the contents of S.R. 1724 and Rost 1745, for I have not managed to see a copy of the latter.

proof: In 1786 Johann Tobias Mayer – to whom I also return – published a book stating that it was a revised version of the "S.R." book from 1724 and that the authors were Schübler and Rost (Mayer 1786, title page and iv).

The section on perspective in *Mathematischer Lust und Nutzgarten* accommodates a nicely illustrated presentation of Desargues's method as described by Bosse (figure XI.11). It also contains references to Niceron and Laraisse.

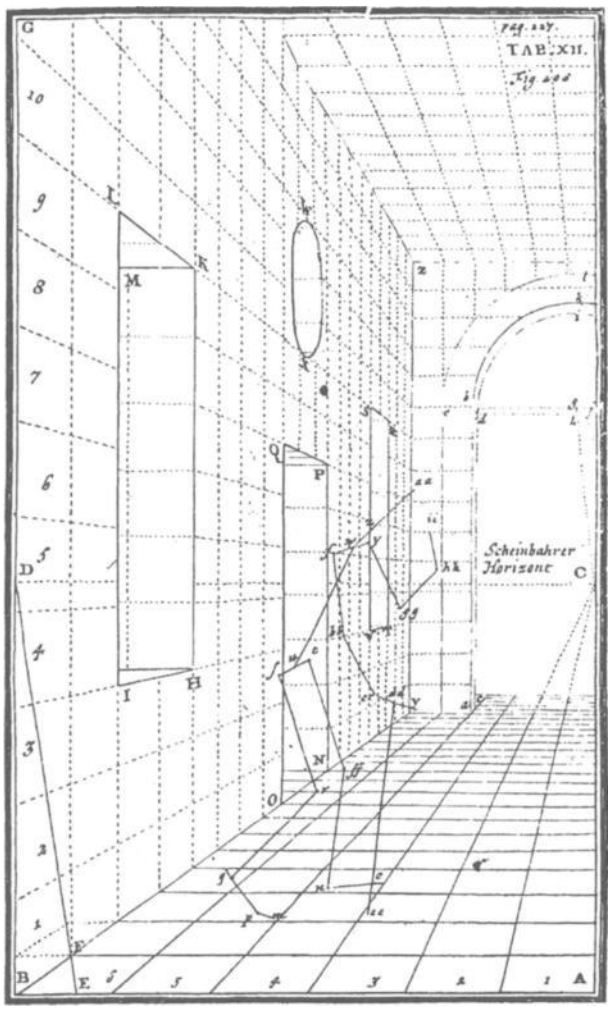

FIGURE XI.11. A perspectival coordinate system. Figure 208 in Schübler & Rost 1724.